📘 MODULE 44: SPLUNK INTERVIEW PREPARATION

How to use this: Read each question, say your answer aloud first, then compare it with the ideal answer. Interviewers care about structure and practical reasoning, not memorized lines. Details like defaults and menu names vary by version, so check the docs for what your target employer runs.

Levels: L1 = entry SOC analyst, L2 = senior analyst or junior admin.

SPL INTERVIEW QUESTIONS

Q1. What is the difference between `stats`, `eventstats`, and `streamstats`?

- **Level:** L1/L2
- **Ideal answer:** `stats` collapses events into summary rows. `eventstats` computes the aggregate and adds it to every original event. `streamstats` computes running values event by event, in order.
- **Why asked:** Tests whether you understand how data changes shape in a pipeline.
- **Common mistake:** Saying `eventstats` "also groups". It doesn't remove events.
- **Follow-up:** How would you find "5 failures in 5 minutes"? (Sort by `_time`, `streamstats time_window=5m count by src_ip`, `where`.)

Q2. `count` vs `dc`?

- **Level:** L1
- **Ideal answer:** `count` counts events, and `dc` counts distinct values. Example: `stats dc(user) by src_ip` finds one IP trying many usernames (password spraying).
- **Why asked:** It is a quick check that you can apply `stats` to security logic.
- **Common mistake:** Using `count` when asking "how many different users".
- **Follow-up:** How do you list those users? (`values(user)`.)

Q3. `search` vs `where`?

- **Level:** L1/L2
- **Ideal answer:** `search` does keyword and field matching with simple syntax. `where` evaluates expressions (comparisons, functions like `like()` and `match()`), and is case-sensitive.
- **Why asked:** Many candidates use them interchangeably.
- **Common mistake:** Using `=` for string comparison inside `eval` or `if()`. Use `==`.
- **Follow-up:** Why does `field!=value` differ from `NOT field=value`? (`!=` excludes events where the field is missing, `NOT` keeps them.)

Q4. Why avoid `join`, and what do you use instead?

- **Level:** L2
- **Ideal answer:** `join` runs a subsearch with limits (rows and time), is slow, and can silently truncate. Prefer one base search covering both datasets with `stats` (or a lookup).
- **Why asked:** Shows performance awareness.
- **Common mistake:** Saying "join is bad" with no alternative.

- **Follow-up:** When is `transaction` justified? (Start/end markers or gap-based sessions.)

Q5. Write a search for the top 10 source IPs by failed logins.

- **Level:** L1
- **Ideal answer:** `index=linux_auth sourcetype=linux_secure "Failed password" | rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})" | top limit=10 src_ip` (or `stats count by src_ip | sort - count | head 10`).
- **Why asked:** A real SOC task, and it shows whether you filter first.
- **Common mistake:** Starting with `index=*` and no time range.
- **Follow-up:** How would you do it from the CIM data model? (`tstats` on Authentication.)

Q6. What does `rex` do and what is a capture group?

- **Level:** L1/L2
- **Ideal answer:** `rex` extracts fields at search time with regex. A named capture group `(?<name>...)` becomes the field.
- **Why asked:** Regex is daily work.
- **Common mistake:** Forgetting `?<name>`, so no field is created.
- **Follow-up:** `rex` vs `regex`? (`regex` filters events and doesn't extract.)

SPLUNK ADMIN INTERVIEW QUESTIONS

Q7. How is Splunk licensed?

- **Level:** L1/L2
- **Ideal answer:** By daily indexed data volume (GB/day). Events dropped to `nullQueue` before indexing don't count.
- **Why asked:** Basic admin knowledge.
- **Common mistake:** Saying by users or searches.
- **Follow-up:** What happens on a license violation? (Warnings, and on some license types search restrictions after repeated violations. Verify per license type.)

Q8. A user says "I can't see my index." What do you check?

- **Level:** L2
- **Ideal answer:** The role's allowed indexes (`srchIndexesAllowed`), then the default search indexes, whether the index exists and has data, and the time range.
- **Why asked:** Roles and access are daily admin tasks.
- **Common mistake:** Only checking whether the data exists.
- **Follow-up:** Where is it defined? (`authorize.conf`, or Settings → Roles.)

Q9. What is an orphaned knowledge object?

- **Level:** L2
- **Ideal answer:** An object, often a scheduled search, whose owner no longer exists. It can stop running until an admin reassigns it.
- **Why asked:** A real operational problem after staff leave.
- **Common mistake:** Never having heard of it.

- **Follow-up:** How do you find them? (Settings → All configurations, filtered by orphaned.)

SPLUNK ARCHITECTURE QUESTIONS

Q10. Explain Splunk architecture.

- **Level:** L1
- **Ideal answer:** Forwarders collect data and send it to indexers, which parse and store it in indexes. Search heads distribute searches to indexers and merge results. A deployment server manages forwarder configuration, and a license manager tracks volume.
- **Why asked:** The first question in many interviews.
- **Common mistake:** Skipping the search head or confusing the indexer and index.
- **Follow-up:** Which component is not in the data path? (Deployment server.)

Q11. UF vs HF?

- **Level:** L1/L2
- **Ideal answer:** The UF is a lightweight agent with no UI and minimal processing. The HF is a full Splunk instance that can parse, filter, mask, route, and run API add-ons.
- **Why asked:** Placement of config depends on it.
- **Common mistake:** Saying the UF indexes data.
- **Follow-up:** Where do you put props/transforms if an HF is in the path? (On the HF, because cooked data isn't re-parsed.)

Q12. How does a distributed search work?

- **Level:** L2
- **Ideal answer:** The search head sends the search (plus the knowledge bundle) to indexers. Indexers search their own buckets in parallel and do streaming and partial work. The search head merges and finishes transforming commands.
- **Why asked:** Tests whether you know why command order matters.
- **Common mistake:** Thinking the search head reads all data itself.
- **Follow-up:** Generating vs streaming vs transforming commands?

SPLUNK TROUBLESHOOTING QUESTIONS

Q13. A UF is running but no data appears. Walk me through it.

- **Level:** L1/L2 (most important question)
- **Ideal answer:** Follow the path. Is the source producing data? Is the input correct and the file readable? `splunk list forward-server` (Active?). `nc -zv indexer 9997`. UF `splunkd.log`. On the indexer, `tcpin_connections` in `_internal`. Does the index exist on the indexer? Then the time range and role permissions.
- **Why asked:** Tests systematic thinking.
- **Common mistake:** Randomly restarting things.
- **Follow-up:** The forwarder shows Active but there is still no data?

Q14. `index=main` works but `index=web` returns nothing.

- **Level:** L1
- **Ideal answer:** Index not created, no data sent to it, wrong time range, role lacks access, or typo.
- **Why asked:** Basic troubleshooting.
- **Common mistake:** Only one cause given.
- **Follow-up:** How do you list indexes with data? (`| tstats count WHERE index=* BY index`.)

Q15. Your configuration change isn't taking effect.

- **Level:** L2
- **Ideal answer:** `btool --debug` to see the winning file. Check `local` vs `default`, the right app and tier, restart or refresh, typos (`btool check`), and whether you are testing with old events.
- **Why asked:** Shows config-management discipline.
- **Common mistake:** Editing `default`.
- **Follow-up:** Why don't index-time changes fix old events?

FORWARDER QUESTIONS

Q16. Which files configure a UF?

- **Level:** L1
- **Ideal answer:** `inputs.conf` (what to collect) and `outputs.conf` (where to send).
- **Why asked:** Basic check.
- **Common mistake:** Forgetting `outputs.conf`.
- **Follow-up:** How do you manage 500 UFs? (Deployment server with server classes.)

Q17. What is the fishbucket?

- **Level:** L2
- **Ideal answer:** Splunk's record of how far it has read each monitored file, so it doesn't re-index data.
- **Why asked:** Explains duplicate or missing data.
- **Common mistake:** Never having heard of it.
- **Follow-up:** Why might a recreated file not be re-indexed?

Q18. Why not receive UDP syslog directly in Splunk in production?

- **Level:** L2
- **Ideal answer:** UDP data is lost when Splunk is down or restarting. Best practice is a syslog server writing files with a UF monitoring them.
- **Why asked:** Tests design maturity.
- **Common mistake:** Saying "UDP is insecure".
- **Follow-up:** How do you set `host` for network inputs? (`connection_host`.)

INDEX QUESTIONS

Q19. Why have multiple indexes?

- **Level:** L1
- **Ideal answer:** Access control, retention differences, performance, and organization.
- **Why asked:** Design reasoning.
- **Common mistake:** Saying only "to organize".
- **Follow-up:** What settings control retention? (`frozenTimePeriodInSecs`, `maxTotalDataSizeMB`.)

Q20. Describe the bucket lifecycle.

- **Level:** L2
- **Ideal answer:** Hot (being written) → warm (closed) → cold (older storage) → frozen (deleted or archived) → thawed (restored manually).
- **Why asked:** Core admin knowledge.
- **Common mistake:** Confusing frozen with thawed.
- **Follow-up:** What happens to events sent to a nonexistent index? (Dropped, with a log warning.)

SOURCETYPE QUESTIONS

Q21. Source vs sourcetype vs host?

- **Level:** L1
- **Ideal answer:** Host is the machine, source is the origin (file path or port), sourcetype is the data format that drives parsing.
- **Why asked:** Foundation.
- **Common mistake:** Saying sourcetype is the OS.
- **Follow-up:** Does changing the sourcetype fix old events? (No.)

Q22. What symptoms indicate a wrong sourcetype?

- **Level:** L2
- **Ideal answer:** Bad event breaking, wrong timestamps, missing fields, and data not appearing in CIM models.
- **Why asked:** Links onboarding to detection quality.
- **Common mistake:** Not connecting it to CIM.
- **Follow-up:** How do you set it explicitly? (`inputs.conf` or props rules.)

PROPS/TRANSFORMS QUESTIONS

Q23. props.conf vs transforms.conf?

- **Level:** L2
- **Ideal answer:** `props.conf` attaches settings to a sourcetype, source, or host. `transforms.conf` holds reusable regex rules that props calls by name.
- **Why asked:** Data onboarding core.
- **Common mistake:** Using the wrong transform name in props.
- **Follow-up:** What do `SOURCE_KEY`, `DEST_KEY`, `FORMAT` do?

Q24. How do you drop events before indexing?

- **Level:** L2
- **Ideal answer:** A transform with `DEST_KEY = queue` and `FORMAT = nullQueue`, called from props, on the first full instance (HF or indexer). Dropped events don't count against the license.
- **Why asked:** Cost control.
- **Common mistake:** Putting it on the UF.
- **Follow-up:** How do you keep only certain events? (Drop all, then re-queue matches with `indexQueue`, in order.)

Q25. How do you mask sensitive data?

- **Level:** L2
- **Ideal answer:** `SEDCMD-` in props (or a transform), at index time on the parsing tier. It only affects new events.
- **Why asked:** Compliance.
- **Common mistake:** Using search-time `rex mode=sed` and thinking the stored data is changed.
- **Follow-up:** Why is search-time masking not enough?

DASHBOARD QUESTIONS

Q26. How do you speed up a slow dashboard?

- **Level:** L2
- **Ideal answer:** Base search with post-process panels, `tstats` or accelerated data models, a narrower default time, scheduled or saved-search panels, and no `index=*`.
- **Why asked:** Real-world pain point.
- **Common mistake:** Saying "buy more hardware".
- **Follow-up:** How does a base search work?

Q27. What are tokens?

- **Level:** L2
- **Ideal answer:** Variables set by inputs or drilldowns and substituted as `$name$` in searches.
- **Why asked:** Tests whether you've built interactive dashboards.
- **Common mistake:** Forgetting default values, causing "waiting for input".
- **Follow-up:** How does drilldown set a token? (`<set token="x">$row.field$</set>`.)

ALERT QUESTIONS

Q28. Report vs alert?

- **Level:** L1
- **Ideal answer:** Both are saved searches. A report displays results, and an alert runs on a schedule (or real time) and triggers an action when a condition is met.
- **Why asked:** Basic vocabulary.
- **Common mistake:** Calling a report a document.

- **Follow-up:** What is throttling?

Q29. How do you reduce alert fatigue?

- **Level:** L2
- **Ideal answer:** Throttle by entity, tune thresholds, use allowlists via lookups, group by entity, and use risk-based alerting.
- **Why asked:** The key SOC pain point.
- **Common mistake:** "Turn off noisy alerts".
- **Follow-up:** Why avoid real-time alerts?

Q30. An alert didn't fire. What do you check?

- **Level:** L2
- **Ideal answer:** Run the search manually over the same window. Check the schedule and enabled state, scheduler logs (`status=skipped`), trigger condition, throttle, email or action settings, and data latency.
- **Why asked:** Troubleshooting depth.
- **Common mistake:** Skipping data latency.
- **Follow-up:** How do you compensate for late events? (Wider window, or `_index_earliest`.)

CIM QUESTIONS

Q31. What is CIM and why does it matter?

- **Level:** L2
- **Ideal answer:** A standard schema of field names and data models, so detections and dashboards work across vendors. ES depends on it.
- **Why asked:** Core for SOC content.
- **Common mistake:** Confusing the CIM app with add-ons.
- **Follow-up:** Add-on vs CIM app? (The add-on maps a vendor's data, and the CIM app defines the models.)

Q32. How do you make a new source CIM-compliant?

- **Level:** L2
- **Ideal answer:** Correct sourcetype → extractions, aliases, calculated fields → eventtype → tag → verify it appears in the data model.
- **Why asked:** Practical onboarding.
- **Common mistake:** Forgetting tags or the index allowlist.
- **Follow-up:** Alias vs calculated field?

DATA MODEL QUESTIONS

Q33. What is a data model and what is acceleration?

- **Level:** L2
- **Ideal answer:** A hierarchical definition of datasets (constraints and fields). It stores no data. Acceleration builds tsidx summaries so `tstats` is fast.

- **Why asked:** Needed for ES-style work.
- **Common mistake:** Saying a data model stores events.
- **Follow-up:** Root vs child dataset?

Q34. Why does `tstats` on a data model return nothing?

- **Level:** L2
- **Ideal answer:** No acceleration with `summariesonly=true`, the constraint matches nothing, no field prefix, data not tagged, or the index is not in the model's allowed indexes.
- **Why asked:** Troubleshooting.
- **Common mistake:** Not knowing about `summariesonly`.
- **Follow-up:** How do you test the model? (`| datamodel ... search`.)

TSTATS QUESTIONS

Q35. Why is `tstats` fast?

- **Level:** L2
- **Ideal answer:** It reads tsidx index files, not raw events, and does no search-time extraction. Limited to indexed fields and accelerated data model fields.
- **Why asked:** I asked about this in your baseline, and interviewers commonly do too.
- **Common mistake:** Trying `tstats` on a `rex` field.
- **Follow-up:** Give a use case. (Silent hosts, ingestion volume, CIM detections.)

Q36. `stats` vs `tstats`?

- **Level:** L2
- **Ideal answer:** `stats` is flexible and works on raw events and any field. `tstats` is much faster but limited to indexed or accelerated fields, and is a generating command.
- **Why asked:** Judgment, not just syntax.
- **Common mistake:** "tstats is always better".
- **Follow-up:** What does `summariesonly` do?

ENTERPRISE SECURITY QUESTIONS

Q37. What is Splunk ES and how does it differ from Enterprise?

- **Level:** L1/L2
- **Ideal answer:** ES is a paid app on top of Splunk Enterprise that adds SIEM workflow: correlation searches, notables, Incident Review, risk-based alerting, asset/identity and threat intel frameworks.
- **Why asked:** Almost guaranteed in a SOC interview.
- **Common mistake:** Treating ES as a separate platform.
- **Follow-up:** Does ES require CIM? (Practically yes.)

Q38. How would you investigate a notable event?

- **Level:** L1

- **Ideal answer:** Open it in Incident Review, assign it, read the description, urgency and risk, use the drilldown to see raw events, check asset, identity and threat intel context, decide true or false positive, document, then escalate or close with a disposition.
- **Why asked:** Mirrors the daily job.
- **Common mistake:** Closing without documenting.
- **Follow-up:** How is urgency calculated? (Severity combined with asset/identity priority.)

Q39. What is risk-based alerting?

- **Level:** L2
- **Ideal answer:** Detections add risk scores to entities (users, systems). An alert fires when an entity's cumulative risk crosses a threshold, reducing noisy single-signal alerts.
- **Why asked:** A modern ES concept.
- **Common mistake:** Confusing risk score with severity.
- **Follow-up:** What is a risk object?